CoEvolution

# A Comprehensive Trustworthy Framework for Connected Machine Learning and Secure Interconnected AI Solutions

*Work Package WP1*

**Project Management**

*Deliverable D1.1*

**Project processes for day-to-day operations and quality management**

| | |
|---|---|
| *Project Acronym:* | CoEvolution |
| *Grant agreement No:* | 101168560 |
| *Funding Scheme:* | HORIZON Research and Innovation Actions |
| *Call:* | HORIZON-CL3-2023-CS-01 |
| *Call Topic:* | HORIZON-CL3-2023-CS-01-03 - Security of robust AI systems |
| *Project Start:* | 1 November 2024 |
| *Project Duration:* | 36 months |
| *Lead Participant:* | ICCS |

This project has received funding from the European Union's Horizon Europe Framework Programme (HORIZON) under Grant Agreement No. 101168560.

## Deliverable Information

| Deliverable Administrative Information | |
|---|---|
| Project Acronym: | CoEvolution |
| Project Number: | 101168560 |
| Deliverable Number: | D1.1 |
| Deliverable Full Title: | Project processes for day-to-day operations and quality management |
| Deliverable Short Title: | Project Manual |
| Document Identifier: | CoEvolution-D11-ProjectManual-submitted |
| Lead Participant: | ICCS |
| Version: | v1.0 |
| Due Date: | 31/01/2025 |
| Submission Date: | 30/01/2025 |
| Dissemination Level: | PU |
| Deliverable Type: | Report |
| Lead Editor(s): | ICCS |
| Author(s): | ICCS |
| Reviewers: | ISI, AEGIS |
| Keywords: | Project management procedures, Communication, Reporting, Quality plans, Risk management |
| Status: | ( ) draft, ( ) final, ( ✓) submitted |

## Deliverable Revisions

| Date | Version | List of Contributor(s) | Summary of Changes |
|---|---|---|---|
| 01/12/2024 | v0.1 | ICCS | ToC drafted, Input solicited |
| 07/01/2025 | v0.2 | ICCS | Content added in all sections |
| 21/01/2025 | v0.3 | ISI, AEGIS | Draft, internally reviewed |
| 23/01/2025 | v0.4 | ICCS | Submitted the updated version to the reviewers, WPL, SPM, and PC, awaiting their approval |
| 30/01/2025 | v1.0 | ICCS | Final editorial check and submit |

## Consortium

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| 11 | DIINEKES S.I. MONOPROSOPI IDIOTIKI KEFALAIOUCHIKI ETAIREIA | DIE | GREECE |
| 12 | CYBERALYTICS LIMITED | CBRL | CYPRUS |
| 13 | SPHYNX TECHNOLOGY SOLUTIONS AG | STS | SWITZERLAND |

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# List of Abbreviations

| | |
|---|---|
| **EU** | European Union |
| **DoA** | Description of Action |
| **EC** | European Commission |
| **WPs** | Work Packages |
| **PC** | Project Coordinator |
| **PO** | Project Officer |
| **GA** | General Assembly |
| **DPC** | Deputy Project Coordinator |
| **CA** | Consortium Agreement |
| **SPM** | Scientific Project Manager |
| **WPLs** | Work Package Leaders |
| **TLs** | Task Leaders |
| **QAP** | Quality Assurance Plan |
| **DL** | Deliverable Leader |
| **IRs** | Internal Reviewers |
| **LB** | Lead Beneficiary |
| **NDA** | Non-Disclosure Agreement |
| **RMF** | Risk Management Framework |
| **RM** | Risk Manager |

# Executive Summary

This specific deliverable outlines the Project Manual which guides the administration, management, and coordination of the project. It sets out the rules and practices to be followed by all partners while engaging in various aspects of the project. Furthermore, it outlines project coordination, continuous monitoring, reporting procedures, governance structure, contractual obligations, decision-making procedures, and other relevant material. This allows for all partners working on the project to be informed and aligned with the requirements and procedures involved. The described management processes are based on and are in line with the procedures defined and agreed in the Grant Agreement and the Consortium Agreement.

The deliverable is structured as follows:

Section 1 presents an Introduction and some basic information about the purpose of the document and its relations with other CoEvolution Work Packages (WPs).

Section 2 presents the designed CoEvolution management structure which encompasses the consortium roles and responsibilities, the decision-making mechanisms, and conflict management procedures.

Section 3 outlines the documents' management procedures such as language requirements, templates, authoring guidelines, exchange methods, document naming and version control, etc.

Section 4 focuses on the procedures to ensure successful communication among project partners including communication tools and the collaborative repository.

Section 5 elaborates on the resource management and project reporting processes.

Section 6 presents the various types of meetings that will be organized during the course of the project.

Section 7 and Section 8 present the Quality Assurance Process and the Risk Management framework that will be followed within CoEvolution.

Finally, Section 9 presents the Conclusion of this deliverable.

# 1 Introduction

## 1.1 Purpose of the document

The D1.1-Project Manual serves as a vital reference for all Project Partners, providing essential guidelines and best practices to be followed in various aspects of the project. It offers a comprehensive Quality Assurance Process to ensure high-quality outcomes, defines roles, duties, and responsibilities, and establishes clear criteria for maintaining quality standards. The project manual emphasizes the importance of adhering to these standards and conducting internal reviews to achieve excellence in project deliverables.

This manual is designed to guide the consortium in the day-to-day activities of the project, facilitating the work of the project coordinator in managing and monitoring project activities. It outlines procedures, structures, and coordination mechanisms necessary for effective project implementation, while also detailing responsibilities for European Union (EU) engagement and interaction. By supporting the efficient administration, procedural and financial management of the project, the manual ensures the timely delivery of results and the achievement of the project's objectives.

## 1.2 Intended readership

As this document is intended to be a source of complete guidance, with standardized procedures for the activities of the project, which will ensure compliance with EU regulations and support effective communication among all partners, it serves as a guide for consortium members to understand their roles, responsibilities, and the overall workflow of the project.

## 1.3 Relationship with other CoEvolution Work Packages and managerial organization

As this document is intended as a guide for the implementation of the CoEvolution project, it indirectly affects all WPs, deliverables, and activities involved in the project. It sets up complete guidelines and procedures for project management, assurance of quality and risk management, which cut across all WPs to ensure consistency among them. The Project Manual helps coordinate and collaborate different WPs during the project to ensure effective communication, alignment, and integration of activities. It describes a quality assurance process and defines roles, responsibilities, and procedures for internal reviews about the preparation of deliverables. As a result, it ensures that all WP deliverables go through a proper evaluation process for quality, accuracy, and compliance with project standards. It assists all WPs in the proactive management of risks and reduction of their possible negative consequences by providing recommendations on risk identification, assessment, and mitigation.

# 2 Management Structure and Decision-Making Procedures

## 2.1 Consortium Roles and Responsibilities

### 2.1.1 Project Coordinator and Deputy Project Coordinator

The **Project Coordinator (PC)** is responsible for the overall coordination and steering of the project. To this end, the PC oversees all activities concerning project planning, monitoring of progress, and delivery of results to ensure that these align with the objectives and plans described in the Description of Action (DoA). In case of deviations, they immediately initiate corrective measures. Moreover, the PC is collaboratively responsible for the knowledge management, innovation activities, and dissemination and exploitation activities of the project. At the same time, the PC acts as a liaison between the Beneficiaries and the European Commission (EC), thus being in charge of all correspondence to and from the EC. This includes leading and preparing periodic reports, organizing review meetings with the Project Officer (PO), and collecting financial statements and audit certificates in the form indicated by the contract. The PC is also responsible for preparing and presenting periodic financial summaries, which enumerate the efforts of resources expended by each of the partners involved. In managing the EU contributions allocated to it, the PC makes sure that funds are distributed to partners according to their allocated efforts. Additionally, the PC convenes and presides over the General Assembly (GA).

The **Deputy Project Coordinator (DPC)** provides vital support to the Project Coordinator in the administrative management of the project. Their primary responsibilities encompass:

- Facilitating effective communication among partners through various tools such as mailing lists, wikis, websites, plenary project meetings, and other methods to ensure efficient collaboration.
- Conducting day-to-day monitoring of all project activities.
- Monitoring compliance by the partners with their obligations under this Consortium Agreement (CA) and the Grant Agreement.
- Organizing project meetings, including scheduling, preparing agendas, taking minutes, and managing related tasks.
- Coordinating the preparation of the periodic Reports.
- Collecting and submitting project deliverables to the Commission.
- Preparing the CA.
- Managing financial activities, including cost claims, payment distribution, and obtaining audit certificates when necessary.
- Preparing and maintaining an up-to-date project calendar and setting up a secure platform for the exchange of project data and information.
- Providing support for knowledge management.
- Coordinating the preparation of project Amendments.

- Transmitting promptly documents and information connected with the Project to any other partner concerned.

Prof. Konstantinos Tserpes (ICCS) has been assigned as the PC of the project and Dr. Antonios Makris (ICCS) has been assigned as the DPC of the project.

### 2.1.2 Scientific Project Manager

The **Scientific Project Manager (SPM)** plays a significant role in ensuring scientific integrity and technical excellence in the project, alongside successful execution. At the same time, SPM acts as the principal deputy to the PC and the DPC for all non-administrative matters, running the project in their absence or representing the project if required.

**Key Responsibilities**

**Overall Technical Management and Coordination:**

- Overall technical management of the project.
- Coordinate and integrate the activities of different WPs to meet the project objectives and schedule.

**Facilitation of Coordination and Collaboration & Progress Monitoring and Risk Management:**

- Monitor the progress of scientific and technical activities.
- Identify deviations or risks and take corrective actions as necessary to keep the project on track.

**Primary Scientific and Technical Communication Point:**

- Represent the project as a first point of contact for most scientific and technical communication.
- Ensure clear and consistent communication among all stakeholders.

**Issue Resolution:**

- Play a critical role in resolving technical or scientific issues that arise during the project.
- Collaborate with Work Package Leaders (WPLs) and other team members in addressing and solving any challenge as soon as possible.

Dr. Apostolos Fournaris (ISI) has been assigned as the SPM of the project.

### 2.1.3 Dissemination Manager

The Dissemination Manager is responsible for the strategic management and execution of communication and dissemination activities within the project. This role involves establishing a framework that enables partners to document their dissemination efforts and assess their impact systematically. By ensuring the implementation of consistent communication initiatives across diverse channels, the Dissemination Manager guarantees that relevant stakeholders are kept informed of the project's progress. Special focus will be given to generating digital content and utilizing social media platforms, to enhance the project's visibility and outreach. Utilizing a range of tools and strategies, the Dissemination Manager ensures impactful communication with the target audience, fostering engagement and maximizing the project's reach.

Ms. Jihane Najar (AEGIS) has been assigned as the Dissemination Manager of the project.

### 2.1.4 Risk Manager

The Risk Manager (RM) is responsible for overseeing the project's entire risk management process, ensuring effective identification, monitoring, and control of risks across all activities. This includes developing and executing the project's risk management action plan. While the RM leads these efforts, active participation from all partners is crucial, with WPLs playing a key role in addressing risks within their specific Work Packages.

Prof. Konstantinos Tserpes (ICCS) has been assigned as the Risk Manager of the project.

## 2.2 Management Bodies

### 2.2.1 General Assembly

The GA shall be composed of one representative from each partner of the Consortium. The GA forms the governing body that oversees the project, and is chaired by the PC. The GA provides a forum to discuss project administrative and strategic management issues and review progress at the dissemination and exploitation levels. This is also responsible for approving significant modifications to project plans, readjustments of efforts, addressing budget-related issues, and considering new partners joining. The evolved changes in the Project Work Plan, the ways of responding to emerging difficulties, or force majeure circumstances will be discussed during the GA. Each partner has one vote in GA meetings. The GA shall not deliberate and decide validly in meetings unless two-thirds (2/3) of its members are present or represented (quorum). Decisions, in general, shall be adopted by a majority of two-thirds (2/3) of the votes cast. Table 1 presents the GA members of the project.

### 2.2.2 Work Package Leaders and Task Leaders

Work Package Leaders (WPLs) are appointed by the partner responsible for their respective Work Package(s) (WPs). Their primary duty is to oversee daily activities within their WP and through close collaboration with Task Leaders (TLs) to ensure effective communication and collaboration. WPLs plan and monitor WP activities, ensuring deliverables are completed on schedule. In their role as WPLs, they are responsible for organizing meetings to facilitate technical discussions. The frequency of these meetings will be determined through discussions between the WPL and contributors, depending on the needs and complexity of the workload.

WPLs are responsible for coordinating interactions and collaborations with other WPs, ensuring effective communication both internally and between different WPs. They submit progress reports and address any concerns.

In summary:

- Coordinate, monitor, and manage activities of the WP ensuring the timely achievement of objectives & milestones.

*Table 1: CoEvolution General Assembly members*

| No | Participant organisation name | Member | Alternate Member |
|---|---|---|---|
| 1 | EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON | Konstantinos Tserpes | Antonis Makris |
| 2 | ATHINA-EREVNITIKO KENTRO KAINOTOMIAS STIS TECHNOLOGIES TIS PLIROFORIAS, TON EPIKOINONION KAI TIS GNOSIS | Apostolos Fournaris | Evangelos Haleplidis |
| 3 | UNIVERSITA DEGLI STUDI DI CAGLIARI | Battista Biggio | Maura Pintor |
| 4 | SIEMENS AKTIENGESELLSCHAFT | Arne Broering | Marco Caselli |
| 5 | AEGIS IT RESEARCH GMBH | Nikolaos Gkatzios | Stella Markopoulou |
| 6 | TELEFONICA INNOVACION DIGITAL SL | Ioannis Arapakis | Carlos Segura |
| 7 | UNIVERSITA DI PISA | Antonio Carta | Davide Bacciu |
| 8 | SECURITY LABS CONSULTING LIMITED | Panos Karantzias | Christos Papadopoulos |
| 9 | AVL LIST GMBH | David Lenk | Bernhard Peischl |
| 10 | AVISENSE.AI TECHNOVLASTOS IKE | Aris Lalos | Theodoros Panagiotakopoulos |
| 11 | DIINEKES S.I. MONOPROSOPI IDIOTIKI KEFALAIOUCHIKI ETAIREIA | Soritis Ioannidis | - |
| 12 | CYBERALYTICS LIMITED | Andreas Miaoudakis | Olympia Giannakopoulou |
| 13 | SPHYNX TECHNOLOGY SOLUTIONS AG | Michalis Smyrlis | Ioannis Arakas |

- Prepare the deliverables required for the WP and support the preparation of overall project management reports.
- Maintain regular communication with the PC, DPC and the SPM, through regular meetings or conference calls, and organize technical meetings or calls with WP contributors.
- Monitor times, costs, and resources, and promptly notify the PC, DPC and the SPM of any deviations.
- Facilitate the flow of information to other WPLs.

Table 2 below includes the list of CoEvolution's WPLs.

### 2.2.3 Task Leaders

The respective WPL coordinates Task Leaders (TLs). The primary role of TLs is the management of day-to-day activities within the assigned task, ensuring effective communication among participants, planning, and monitoring the activities of functions. TLs ensure that the outcomes

*Table 2: CoEvolution Work Package Leaders*

| No | Name | Leader | Organisation |
|---|---|---|---|
| WP1 | Project Management | Konstantinos Tserpes | ICCS |
| WP2 | Dissemination, Exploitation, Sustainability | Andreas Miaoudakis | CBRL |
| WP3 | Specification / Requirements and CoEvolution Architecture | Apostolos Fournaris | ISI |
| WP4 | AI model STR Assessment, Enhancement and Context Awareness technology enablers | Ioannis Arapakis | TID |
| WP5 | CoEvolution AI model lifecycle STR Security Assurance and Monitoring | Sotiris Ioannidis | DIE |
| WP6 | Integration, evaluation and project outcomes | Aris Lalos | AVI |

of the tasks - deliverables - are completed on time. While tasks require contribution from other partners, the primary responsibility for managing the assignment belongs to the TL.

## 2.3 Decision-Making Mechanisms

Technical decisions at the WP level are mainly a matter of the corresponding WPL. However, if a technical decision will have repercussions on the work of others, the WPL should first consult the PC, DPC and the SPM, as well as all other WPLs. In such a way, interdependencies among WPs will be managed effectively.

Project-Level Decision-Making: The PC, DPC and the SPM, take technical decisions at the project level. This has to be done coherently, respecting the general goals and strategies of the project. Decisions made by the GA are binding on the whole project to avoid different directions and non-standard implementations and protocols.

**Decision-Making Responsibilities:**

- Work Package Leaders (WPLs): Make technical decisions within a given WP; consult with PC, DPC, SPM, and other WPLs when the decision affects more than one WP.
- Project Coordinator (PC): Ensures overall supervision and inter-WPs coordination.
- Scientific Project Manager (SPM): Ensures that the project is guided toward achieving scientific excellence.
- General Assembly (GA): Holds the ultimate decision-making power, with decisions binding on the whole project.

## 2.4 Problem Handling and Corrective Actions in CoEvolution Project

**Problem Handling and Corrective Actions:** The consortium approach for problem handling and corrective action would be oriented towards prevention. Once an existing or potential problem is identified, the same shall be reported in writing to the PC immediately, which shall be addressed at the earliest detectable stage and the lowest level with less impact on project progress. Each consortium partner shall be responsible and accountable for its obligations concerning the Project towards the EU Contract. Any misperformance or underperformance by any of the partners will be instantly documented.

- **Efforts for Immediate Resolution:**
  - The PC will exhaust all possible means to solve issues.
  - In case of need, the PC shall send an official e-mail to the concerned partner, thus notifying the GA and the PO in charge of the Project.
- **Consensus Decision-Making:**
  - The GA shall make critical technical and other decisions through consensus.
- **Negotiation and Voting:**
  - First of all, disputes will be resolved through negotiations.
  - In case consensus is not possible, a majority vote shall be applied. All GA members will have one vote; if there is a tie, a PC member will provide the tiebreaker vote.
- **Escalation for Persistent Disputes:**
  - In the case of persistent or high-impact disputes jeopardising the continuation of the project, the consortium shall contact the PO, request advice from outside, and organize a meeting with all consortium members.

Responsibilities:

PC: Problem solving in the short term, official correspondence, arrangement of GA meetings, and casting tie-breaking votes.
GA: Consensus decision-making, handling significant conflicts, and voting on resolutions.

Following this structured approach, the consortium can handle problems in the most effective possible manner with least disturbance and allowing the project to progress.

# 3 Document Management

Microsoft SharePoint has been chosen as the designated software for the document repository, which will host all essential files throughout the project's lifecycle. Specifically, this includes Deliverables, Administrative Documents, Reports, Dissemination Material, Cost Claims, Meeting/Teleconference Minutes & Action Items, Annual Project Reports, Contractual Documentation, Technical Reports, Technical Papers, and Market Studies.

Link to the CoEvolution SharePoint: https://ntuagr.sharepoint.com/teams/CoEvolution-EU

## 3.1 Document Template

The PC, in collaboration with the DPC, has developed all required templates. Each CoEvolution deliverable will conform to a uniform Overleaf template (LaTeX), guaranteeing both consistency and a cohesive visual presentation. This template is publicly accessible and available within the CoEvolution GitHub organization at: https://github.com/CoEvolution-Project-EU. It incorporates a suite of quality control measures designed to elevate the standard of project deliverables and includes features that enhance document quality. Furthermore, all presentations associated with the project, whether for external events or internal meetings, will utilize the designated PowerPoint template, which is available in the "Templates-Logos" directory of the CoEvolution Microsoft SharePoint repository. Additionally, partners are required to utilize the provided reporting template to document their quarterly progress.

## 3.2 Guidelines for Authoring Deliverable Submissions

In this section, we will describe the guidelines for authoring deliverable submissions.

**Template**

The deliverable must adhere to a universal Overleaf template (LaTeX), ensuring consistency in formatting, structure, and presentation across all documents. This template serves as the foundation for maintaining professional standards and uniformity in documentation.

**Change Log**

A "Deliverable Revisions" table (change log) must be included in the deliverable to document versioning and contribution history. This table tracks all changes made to the document, detailing revisions, updates, and the progress of the Quality Assurance Plan (QAP) (described in Section 7). This systematic tracking is essential for the transparency, accountability, and traceability of the document's development.

**Executive Summary**

An Executive Summary, limited to one page, is required to provide a concise overview of the deliverable's findings. This summary should highlight the scientific, technical, or business value of the document, offering stakeholders a quick yet comprehensive understanding of its significance and key outcomes.

**Introduction**

The introduction should be brief, spanning 1-2 pages, and clearly articulate the purpose and scope of the deliverable. It must explain its relationship to other WPs or Deliverables, contribute to the overall WP and project objectives, and provide an outline of the document. This section sets the context and framework for the reader.

**Conclusion**

The deliverable should conclude with a section that summarizes the findings and outlines plans for future work. This conclusion provides closure and direction, highlighting the next steps and potential avenues for continued research or development.

**File Naming**

Given the project's extensive documentation, an agreed file naming format is essential for organization and retrieval. Detailed guidelines for this format are provided in Section 3.4, ensuring systematic and efficient file management.

**Figures, Tables, and Terminology**

Figures: All figures must include relevant captions and be introduced within the text.
Tables: Similar to figures, tables must also have captions and be introduced within the text, with captions placed above the tables.
Terminology: A comprehensive list of terminology used in the deliverable must be included, ensuring clarity and consistency in language and definitions.

**Language**

UK English: The deliverable must consistently use UK English as the official language, checked thoroughly for spelling and grammar to maintain linguistic accuracy and professionalism.

**References**

IEEE Style: References should adhere to the IEEE Style. Every reference listed must be cited within the text, and all citations in the text must appear in the reference list, ensuring completeness and accuracy. Plagiarism must be strictly avoided to maintain the integrity of the document.

**Document Formatting**

Headings: All headings should be capitalized without a full stop at the end.
Captions: Table captions should be placed above the table, and figure captions should be placed below the figure, ensuring a standardized format that enhances readability and aesthetic consistency.

**Outputs**

A dedicated section should detail any datasets, code, or other resources being released, if applicable. This section ensures that all supplementary materials are clearly documented and accessible for future use.

**General Guidelines**

The deliverable should be a self-contained document, providing all necessary information within its pages without requiring external references. This ensures that the document is comprehensive, accessible, and understandable as a standalone piece. By adhering to these guidelines, the deliverable will maintain high standards of quality, clarity, and professionalism, effectively communicating its content to all stakeholders.

## 3.3 Document Exchange Methods

The critical approach for document sharing will be through file upload into the CoEvolution Microsoft SharePoint repository. Once a document has been uploaded, the partner is inducted to mail the document title with its link. Alternatively, for documents that are temporary or when uploading is not practical, documents will be exchanged via email.

## 3.4 Document Naming & Version Management

Proper document naming will facilitate the management of the project's technical and administrative resources. CoEvolution will implement a standardised naming convention for all versions of project deliverables so that proper versioning is maintained. A primary naming convention of this nature may include the following elements:

- **Project Acronym:** A project acronym is the standard project abbreviation.
- **Deliverable Identifier:** Indicates the deliverable identifier. Standard code dxx.
- **Deliverable Short Title:** A brief title without spaces.
- **Status Indicators:**
  - **"draft":** The deliverable is currently being drafted.
  - **"final":** The last version after review, updating, and approval by the reviewers, deliverable leader, WPL, SPM, and PC.
  - **"submitted":** The version sent to the European Commission by the PC.
- **Version Number:** Every new revision will increase incrementally the version number. Format will be -vX.X. The last version will be v1.0.

This will create guidelines for the systemic tracking version control of the work throughout the project.

**Internal reference number example of deliverable:**

- CoEvolution-D82-ProjectManual-draft-v0.6.pdf

The Deliverable Leader shall organise the input of all Contributing Partners and assemble the deliverable. Each deliverable shall contain a "Deliverable Revisions" table on its second page with information about versioning and contribution history to keep track of document changes. In the event of possible changes being made, a new version shall be created, together with a short description of the changes done.

## 3.5 Document Processing Software Tools

The following are the recommended tools for handling documents:

**Word Processing:** Microsoft Word version 2010 or above, Mac and PC, Microsoft 365.
**LaTeX Processing:** Overleaf (online)
**Spreadsheet:** Microsoft Excel (2010 or later, both MAC and PC), Microsoft 365.
**Presentation Preparation:** Microsoft PowerPoint; version 2010 or later; Mac and Windows, and Microsoft 365.
**PDF manipulation:** Adobe Acrobat (2010 or later version; Mac and PC).

Lastly, if any partner wants to use any other software, it remains their responsibility that the output produced works with these recommended tools.

# 4 Communication

Effective communication is fundamental for the successful management and execution of collaborative projects. The project's communication infrastructure is designed to be fast, reliable, and easily accessible among partners and external entities.

## 4.1 Communication Infrastructure

Key electronic communication methods include email and web-based platforms. Additionally, a dedicated project website enhances the speed and efficiency of information dissemination.

## 4.2 Primary Communication Channels

The primary means of communication among the consortium partners is the official CoEvolution mailing list. AEGIS and ICCS has established and maintains the following email list:

- **all@coevolution-project.eu**: This list includes all members working on the project and is used for discussing all project-related matters.
- **ga@coevolution-project.eu**: The mailing list of the GA.
- Additional lists will be created within the CoEvolution Project for separate communication.

We chose to have a single general mailing list to ensure everyone is involved and updated on the work taking place in CoEvolution. If a new participant needs to be added to the list, the designated administrators Nikolaos Gkatzios (AEGIS), Antonis Makris (DPC) and Konstantinos Tserpes (PC) should be contacted via email.

Except for the official mailing list, CoEvolution Project utilizes a variety of communication channels to maintain effective internal and external communication.

These channels include:

- **Email:** For regular, asynchronous communication.
- **Slack** / **Microsoft Teams:** For real-time communication and collaboration.
- **Microsoft SharePoint:** For document management and collaborative work.
- **Physical Meetings:** For in-depth knowledge exchange and strategic planning.

## 4.3 Internal Communication

The internal communication of CoEvolution is organised in a fashion that supports complete knowledge transfer and coordination:

- **In-person meetings:** This has begun with a two-day-long kick-off meeting for the establishment of foundational understanding and goals. The aim is to hold three in-person meetings annually to review progress, align on objectives, and address key milestones.
- **Teleconferences:** The frequency of these depends on the relevant work, ensuring flexibility to address progress and decisions effectively.

- **Email Exchanges and SharePoint:** prolonged communication and document sharing.

## 4.4 External Communication

For external communication, the results and achievements about CoEvolution are forwarded through a variety of channels such as:

- **Publications:** Academic and industry-related publications.
- **Conferences and Events:** Participation to present results and network with peers.
- **Connection with related projects, associations of SMEs:** This will enable collaboration and create broader impact.

### 4.4.1 Communication Flow

Communication flow at CoEvolution is designed with systematic information sharing, timely delivery, and avoidance of overload:

- **Work Package (WP) Meetings:** Focused meetings among WP members for specific task coordination.
- **Day-to-Day Communication:** Through calls and emails, supporting ongoing collaborative work.

### 4.4.2 Role of the Project Coordinator (PC)

The PC plays a pivotal role in maintaining communication:

- **Daily Communication:** With all partners so that the same consistency of information could be conveyed across.
- **Partner point-to-point communication:** Direct communication among concerned partners, with the possibility of escalation to the WP Leaders (WPLs) and Taks Leaders (TLs) if necessary. Encourage ad hoc communication to discuss urgent issues and improve overall collaborative efficiency.

## 4.5 Project Collaborative Shared Drive Repository

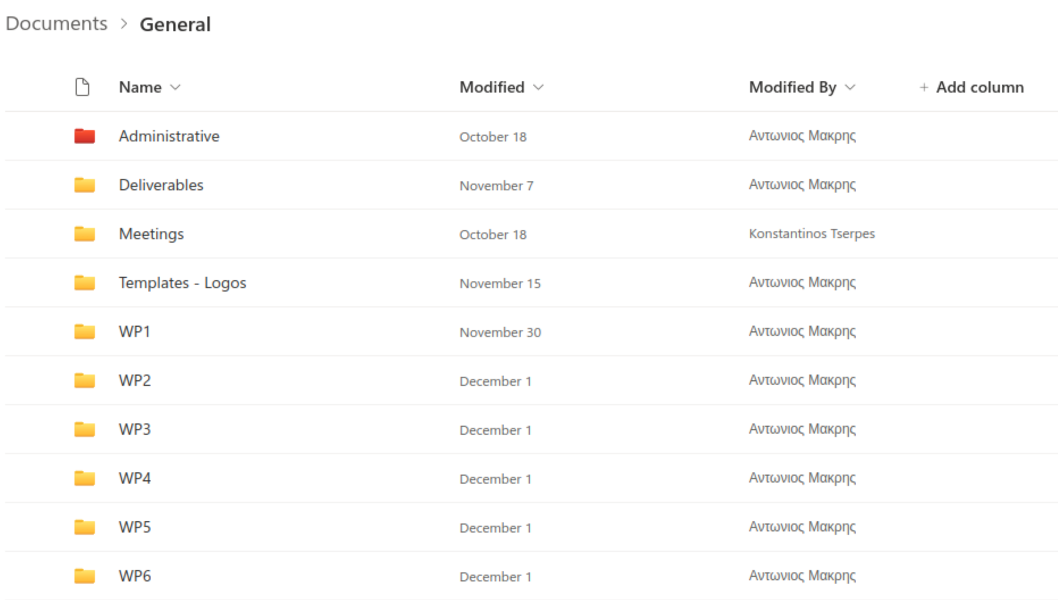
As mentioned in Section 3, a SharePoint repository for information exchange and file sharing among CoEvolution project partners has been established and is referred to as the *CoEvolution-EU repository*. This repository is instrumental in ensuring that teamwork and smooth communication within the project work effectively.

**Repository Administration**

The administration of the CoEvolution repository is managed solely by the PC and the DPC, who are responsible for:

- **Account Creation:** Setting up user accounts for new consortium members.
- **User Authorization:** Granting access to the repository based on requests from project partners.

- **Modifications:** Making any necessary changes to user permissions and repository settings to ensure smooth operation and security.



Documents > General

| Name | Modified | Modified By | + Add column |
|---|---|---|---|
| Administrative | October 18 | Αντωνιος Μακρης | |
| Deliverables | November 7 | Αντωνιος Μακρης | |
| Meetings | October 18 | Konstantinos Tserpes | |
| Templates - Logos | November 15 | Αντωνιος Μακρης | |
| WP1 | November 30 | Αντωνιος Μακρης | |
| WP2 | December 1 | Αντωνιος Μακρης | |
| WP3 | December 1 | Αντωνιος Μακρης | |
| WP4 | December 1 | Αντωνιος Μακρης | |
| WP5 | December 1 | Αντωνιος Μακρης | |
| WP6 | December 1 | Αντωνιος Μακρης | |

*Figure 1: CoEvolution Sharepoint Repository*

Once access is granted, the users shall observe the following guidelines to effectively use the repository:

- **File Sharing:** All project documents, presentations, and data relevant to the project must be placed in the repository for easy access and thus sharing among the partners.
- **Version Control:** Versions of documents are to be kept and traced chronologically for successive modifications and updates; this would prevent conflicts and ensure that all the partners work with the most current information.
- **Information Security:** All uploaded files should demonstrate compliance with the information security policies as required by the project. Sensitive information shall be dealt with using protocols identified by both parties to prevent unauthorised access or leakage of information.
- **Regular Updates:** New information, findings, and contributions shall be posted constantly in the repository to keep all partners updated.

By following these guidelines, the CoEvolution repository shall be a vital tool for collaboration, allowing all partners of the project to locate information and resources that would help in making their contribution to the success of the overall project undertaking. The role of the PC and DPC in running the Management of the Repository shall be most significant in ensuring the integrity and security of the repository and supporting the overall goals of the CoEvolution project.

**Folder Structure and Management**

Centralised repository with an intuitive folder structure, making it easy to navigate and manage documents. The initial structure of the repository, as designed by the PC and the DPC, will be:

- Work Package (WP) folders: Each WP has its folder where all the documents related to this WP are kept, including deliverables and meeting minutes.

  - Customisation by WPLs: While the PC and the DPC set up the general structure, WPLs can further customise their respective WP folders to best suit the needs of their team.
  - Maintenance Responsibility: The maintenance and organization of WP documentation is the responsibility of the respective WP teams.

- Additional Folders
  - Deliverables: contains the final version of all deliverables submitted.
  - Administrative: contains administrative files.
  - Meetings: contains material related to the meetings.
  - Templates - Logos: contains material related to the logos of the project and the templates for the presentations, meeting minutes and quarterly reports.

Table 3 below summarizes the structure of the CoEvolution Sharepoint repository.

*Table 3: CoEvolution Sharepoint Folder structure*

| Folder name | Description |
|---|---|
| Administrative | Administrative Files Subfolders: Boards-Leads, GA, CA, Risks |
| Templates- Logos | Logos of the project and templates for all Consortium members. Subfolders: Logos, Templates |
| Meetings | Material related to the Meetings arranged by the Consortium (agenda, minutes, participant list, member presentations, meeting photos, and other relevant documents) |
| Deliverables | Copies of submitted deliverables |
| WP1 | Materials related to the overall management of the project including deliverables, documentation, QAP, reporting, risk management |
| One folder per WP (WP2-WP6) | Materials specific to that WP including deliverable drafts, Work-In-Progress and contributions of task leaders and partners per task. The WPLs are responsible for keeping these folders up to date. |

# 5 Project Reporting

The reporting procedure aims to evaluate the project's overall performance and provide a summary of its results. Project reporting in CoEvolution is divided into two categories: Periodic Reporting & Continuous Reporting. The procedures for both types of project reporting are described in detail in this section.

## 5.1 Periodic Reporting

Beneficiaries are required to provide regular periodic reports in accordance with the DoA, detailing actions such as deliverables, milestones, outputs/outcomes, critical risks, and their mitigations. According to the GA the first review occurs within two months after the end of the first reporting period (M18) while the second occurs between M36-M38. The internal deadline for submitting the necessary input to the PC is 15 days prior to the review. This ensures that the PC has sufficient time to prepare the project report before the actual project review. A draft version of the report should be delivered to the PO and the reviewers in advance.

## 5.2 Continuous Reporting

The continuous reporting procedure follows a hierarchical approach to ensure that implementation aligns with the work plan & project outputs correspond with the DoA. TLs are responsible for monitoring and reporting the project's progress and possible risks. They report directly to the WPL who in turn report to the GA. Reports may be delivered during WP meetings, Plenary meetings, and/or GA meetings.

### 5.2.1 Technical Progress Reporting

Quarterly, reports should be sent to the PC within ten working days after the end of the technical reporting period, using the specific template. Such quarterly reports should include, but not be limited to, the following:

- Significant achievements per partner,
- Progress per work package,
- Status of deliverables,
- Deviations from the work plan,
- Conferences, Status of publications,
- Status of talks given by the partner,
- Other achievements relating to the project of importance.

### 5.2.2 Financial Reporting

The Cost Claim reports shall be sent to the PC and the WPL within two weeks from the end of the reporting period, using a template that will be provided by the PC. This template is based on the information included in the official report of the European Commission. The PC shall review the resource and budget consumption, providing support when necessary. The Cost Claim Reports shall contain, at a minimum, the following elements:

- Planned & Actual Resources per task per Work Package
- Consumables
- Travel expenses
- Hardware/Software Costs
- Auditing Costs

### 5.2.3 Reporting on Work Package & Task Level

**Work Package Level Reporting**

WPLs are technical development coordination managers of their respective tasks and activities aligned with the work plan. The role of a WPL shall include:

The preparation of internal and external reports, which become essential for the WP Reports, and assists in preparing overall management reports. Ensuring a minimum of one monthly contact with the respective TLs for follow-up on technical progress within the WP and arranging regular meetings with the SPM and the PC to provide information on progress and potentially critical issues or deviations. Keeping minutes of the meetings and extracting a list of action items detailing open issues, the task, deadlines, the partner assigned to each task, a brief description, and status of the issue. The action item list should be maintained in the Microsoft SharePoint repository, accessible to all project partners. Organization of technical presentations of WP activities for proper involvement and visibility of WP contributors. Facilitates horizontal information flow and interaction with other WPLs.

**Task Level Reporting**

Task Leaders (TLs) shall be responsible for overseeing the daily operations within their designated tasks. TLs should attend meetings organised by the WPLs and provide reports on the progress and achievements of their tasks.

# 6 Project Meetings and Reviews

Project meetings are crucial for the effective management of the CoEvolution project, serving as a communication hub for partners, aiding in decision-making, resolving issues, and tracking progress. Different types of meetings will be organized to support these objectives, as detailed in the following sections.

## 6.1 General Assembly Meetings

General Assembly (GA) meetings will be convened to address significant administrative or technical issues within the CoEvolution project as they arise. These meetings have the authority to result in modifications to the project consortium, alterations to project objectives, or even project termination. Initiated and chaired by the PC, participation from each consortium organisation is mandatory.

The PC shall convene ordinary meetings of the GA at least once every six months and shall also convene extraordinary meetings at any time upon written request of any consortium organisation. All partners will receive notifications at least 2 weeks prior of an ordinary meeting and at least one week preceding an extraordinary meeting. Agendas and supporting documents will be distributed to all attendees at least two weeks before the meeting or one week before an extraordinary meeting. The PC is also responsible for consolidating submissions from partners. The PC, or their representative, will record the meeting minutes, which will be issued within ten calendar days of the meeting unless otherwise agreed. To minimise costs and optimise time, GA meetings will be scheduled alongside plenary meetings whenever possible. Each meeting will commence with an agenda agreement and conclude with the setting of the date and location for the next meeting.

## 6.2 Project Plenary Meetings

Plenary meetings are scheduled to occur at regular intervals of four months. However, additional plenary meetings may be convened at any time should a significant technical issue arise. A WPL can also initiate a plenary meeting with prior approval from the PC. Attendance from representatives of all consortium partners is mandatory. These meetings will be initiated by the PC. Scheduled meetings will be announced at least three (3) weeks in advance. In exceptional circumstances where urgent issues arise, notifications for meetings can be issued one week in advance, provided that at least 70% of the participating partners are available. Essential details are outlined below:

- **Agenda and Documentation:**
  - The agenda and supporting documentation will be made available to all attendees at least one week before the meeting. The PC is responsible for issuing these documents.
- **Meeting Minutes:**
  - The PC, or an appointed substitute, if necessary, is responsible for providing the written minutes. In the event that the meeting focuses on a specific work package, the corresponding WPL will be responsible for providing the minutes.

  - Unless otherwise agreed during the meeting, the minutes will be issued within 15 working days after the meeting ends.
  - To ensure brevity and clarity, the minutes will be formatted as action points.
- **Documentation Management:**
  - The list of action points will be maintained in a file on the CoEvolution SharePoint, accessible to all project partners.
- **Meeting Procedures:**
  - Each meeting will begin with an agreement on the agenda and conclude with an agreement on the date and location of the next meeting.

## 6.3 Project Review Preparation Meetings

Project Review Preparation Meetings (RPMs) will be held at least two weeks prior to the official project review sessions. One of the main objectives of these preparatory meetings are to check progress toward the realisation of the project objectives.

Any deviations from the planned progress will be reported to the General Assembly along with a proposed mitigation strategy. The PC will organise these meetings, and attendance by representatives of all participating organisations is mandatory.

**Key Protocols**

**Agenda & Documentation:**

- The PC shall prepare and distribute the agenda of meetings, together with supporting documents about it, at least one week before the RPM.

**Meeting Minutes:**

- The PC, or a designated substitute if necessary, will record and distribute the meeting minutes.
- Minutes will be action-point based to make them concise and easy to read.
- The minutes will be issued within ten working days following the meeting unless another timeline, extended or shortened, is agreed to during the meeting.

**Action Points:**

- All action points will then be summarised in a document kept on CoEvolution SharePoint to ensure the document's availability to all project partners.

## 6.4 Project Review Meetings

Project Review Meetings are an integral part of the life cycle of collaborative projects and shall be organized within 60 days following the end of each reporting period, unless otherwise requested by the funding authority. These full-day events are formal occasions for the consortium to present the progress and achievements of the preceding period to the PO and an independent external review panel.

**Objectives and Structure**

The Project Review Meeting focuses on whether the project is progressing toward its goal, milestones, and deliverables. Essentially, it is a review of work completed or the percentage of work done against the plan, an assessment of how much the project is aligned with the scheduled plan, budget, and quality standards, and an identification of miscellaneous challenges or risks likely to hamper progress.

**Roles and Responsibilities** The PC is instrumental in organising and preparing for these meetings. The role of the PC will be to:

- Objective Setting: The need to clearly set out objectives for the Project Review in order to align stakeholders' expectations.
- Agenda Preparation: Detailed preparation of the agenda, reviewed and approved by the PO. This indicates the schedule, especially the key discussion topics and time allocations expected for presentations and discussions.
- Role Assignment: Delegating specific roles and responsibilities to consortium partners, ensuring that each partner is prepared to present their contributions and answer questions.
- Contribution Call: Collection of input and materials from consortium partners to develop extensive reports and presentations reflecting the status of the project.

**Meeting Preparation**

In preparation for the Project Review Meeting, the PC will be organizing an internal rehearsal on the preceding day(s) between all consortium partners. This session serves several purposes:

- Presentation Practice: Allows partners to practice presentations and be clear and coherent.
- Feedback & Improvement: This phase provides an opportunity to take peers' feedback, identification of gaps, and its adjustment.
- Technical Check: This ensures that everything, from presentation gear to software, works fine to avoid disruptions during the actual review.

**Meeting Execution**

Whereas in the Project Review Meeting, actual detailed updates on the different aspects of the project are presented by the consortium, covering:

- Technical Progress: Summary of work done, results obtained, and any deviations from the work schedule.
- Financial Status: Budget expenditure update, budget-financial management, and any variances therein.
- Risk Management: Identifying and mitigating probability risks encountered during the reporting period.
- Future Plans: Outline of the following steps to be taken, some impending milestones, and which strategic adjustments will occur according to the review feedback.

These presentations are critically reviewed by the PO and independent reviewers, who question and comment on them. Their assessment forms a base to continue either, adjust, or even probably redirect the project's trajectory. Project Review Meetings are an integral part of maintaining transparency, accountability, and efficiency in collaboration within multi-partner projects. It enables the consortium to ensure that with due care in the preparation and conduction of reviews, all stakeholders will have an overview, be engaged, and be aligned with the objectives and progress of the project, hence increasing the chances of successful project outcomes.

## 6.5 Guidelines for Acknowledging EU funding – Disclaimer

The CoEvolution project is dedicated to increasing the visibility and impact of its research through active engagement in conferences, dissemination activities, and standardisation bodies. To ensure appropriate acknowledgment of EU funding and compliance with project guidelines, specific protocols must be adhered to when expenses are incurred using the CoEvolution budget.

The following text should be used:

*This [paper/presentation/work/research/...] [has received funding from/has been supported by] the European Union's Horizon Europe research and innovation actions under grant agreement No 101168560.*

To maintain consistency and avoid potential duplication or conflicting contributions within the CoEvolution project, the following guidelines have been established:

**Documentation Upload to CoEvolution Sharepoint**

All conference presentations, documentation, and other contributions related to the project must be uploaded to the CoEvolution Sharepoint. This centralised repository ensures that all project related materials are accessible to consortium members and can be reviewed and referenced as needed.

**Notification to the Project Coordinator**

To ensure proper coordination and oversight, consortium members must inform the PC of any dissemination or communication activity related to the project and provide a brief report of the activity no later than two weeks after it has occurred. This process ensures that the PC can:

- Maintain a comprehensive and accurate record of dissemination and communication activities for reporting purposes.
- Verify that the activities align with the project's goals and messages.
- Use the information to coordinate future efforts and avoid overlapping or conflicting activities.

**Inclusion of a Disclaimer**

It is advisable to include a disclaimer in every publication or presentation associated with the CoEvolution project. This disclaimer helps clarify the relationship between the authors and the funding bodies, and it protects the project and the funding authorities from any unintended implications. A typical disclaimer text is provided below:

*Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or [name of the granting authority]. Neither the European Union nor the granting authority can be held responsible for them.*

By adhering to these guidelines, consortium members can ensure that all communications related to the CoEvolution project are well-coordinated, clear, and consistent with the project's overall objectives and messages. This contributes to the professional representation of the project and upholds the integrity of the consortium's work.

# 7 Quality Assurance Process

Quality Assurance encompasses the activities and processes established to ensure that the project's products, services, and activities meet the required quality standards. It involves systematic measurement, comparison with standards, process monitoring, and a feedback loop to prevent errors. Quality assurance is based on two principles:

- **"Fit for purpose"** (the product should be suitable for its intended use)
- **"Right first time"** (eliminating mistakes)

This section will describe the Quality Assurance Plan for the CoEvolution project. Its purpose is to define a consistent set of evaluation methods and procedures to be applied throughout the CoEvolution project to ensure quality and conduct evaluation studies systematically. The contents of this document align with the project's Grant Agreement (GA). The Quality Assurance Process (QAP) applies to all project activities, and adherence to the Quality and Assessment Plan is mandatory for all parties involved.

In Table 4, the QAP steps for the deliverable preparation and submission process are described. The deadlines indicated on each step in the "when" column is indicative but strongly recommended in order not to delay the submission of deliverables to the EU Portal.

These deadlines can be further changed if required on a need basis with documented reasons such as scope of deliverables, actual progress, or other unprecedented circumstances. Any modifications to the deadlines must be mutually agreed upon by the responsible authorities (specified in the "who" column) and those overseeing compliance at each step (outlined in the "what" column). Regardless of any adjustments, deliverables must be finalised and submitted to the PC no later than one day before the official submission deadline. Each deliverable should utilise its corresponding workspace in Overleaf, where the Table of Contents, draft versions, clean versions before reviews, reviews, and the final version to be submitted are stored and managed. Overleaf's versioning system ensures that all changes and revisions are securely tracked and accessible. It is the responsibility of the Lead Beneficiary (LB) (also referred as Deliverable Leader (DL)) to provide regular draft versions of the deliverable to keep the other partners up-to-date with progress, potential problems conveyance, and any other relevant information.

Figure 2 illustrates the QAP steps involved in the deliverable preparation process.

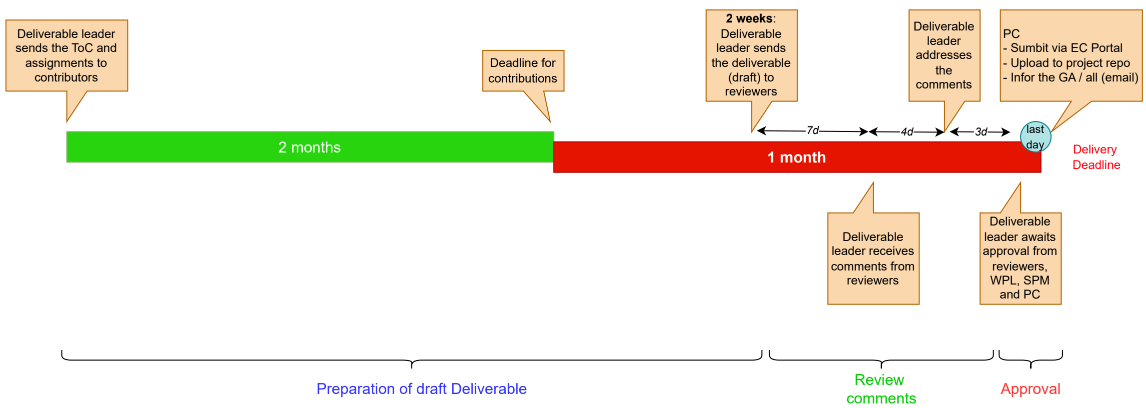



*Figure 2: Deliverable proccess pipeline*

*Table 4: QAP Steps*

| Step | When | Who | What |
|---|---|---|---|
| 1 | 2 months before the submission deadline | DL | The DL prepares the Table of Contents (TOC) in collaboration with the deliverable contributors. A brief explanation of the expected content is included along with writing assignments (indicating the responsible authors for each section), estimated length of each contribution, & writing deadlines (indicating the due dates for each contribution). The DL also requests timely input from the contributors to the deliverable, requiring all contributions to be completed no later than one month before the deliverable submission deadline. |
| 2 | 2 weeks before the submission deadline | DL | The DL submits the initial draft of the deliverable to the Internal Reviewers (IRs). DL request their review within one week, while copying WPL, SPM, and PC. |
| 3 | 1 week before the submission deadline | IR | The IRs provide their feedback to the DL. Feedback is provided through comments and tracked changes in the online Overleaf deliverable. |
| 4 | 3 days before the submission deadline | DL, IRs, WPL, SPM, PC | The DL addresses the comments from the IRs and submits the updated version to the reviewers, WPL, SPM, and PC, awaiting their approval. |
| 5 | 1 day before the submission deadline | DL, PC | During the final step, the DL submits the final version to the PC, who finalises the approval and submits the final PDF file to the EU Portal. |

## 7.1 Quality Assurance Objectives

The primary objective of the Quality Management Plan is the establishment of a framework for the quality assurance procedures that will be implemented throughout the duration of the project.

The objectives could be summarised to the following:

- Standardisation of processes for communication among partners, document sharing, review procedures, and the delivery of both tangible and intangible project results.
- Specification of tools & criteria for evaluating the quality of project outcomes.
- Evaluation of deliverables based on the established criteria.
- Planning and implementing corrective actions to address any deviations in project outcomes concerning time, quality, and cost.

## 7.2 QA Roles & Responsibilities

This section outlines the Quality Assurance (QA) roles and responsibilities assigned to consortium partners, which will guide the implementation of the project's QAP:

- The **LB** (also referred as Deliverable Leader (DL)) bears the primary responsibility for the comprehensive preparation of the deliverable. Furthermore, the LB is accountable for the deliverable's overall quality and for ensuring that all feedback from the Internal Reviewers (IRs) is duly addressed. Each contributor to the deliverable is responsible for the quality of their respective contributions and for addressing the comments made by the IRs.
- **IRs** are members of the CoEvolution consortium, designated by the PC to evaluate specific deliverables. Their duty is to execute the internal review process for the deliverable.
- The **SPM** is responsible for steering the project towards scientific excellence.
- The **PC** is responsible for the Quality Assurance Management of the project, approving the submission of deliverables to the EU. Additionally, the PC is tasked with making decisions and applying policies to ensure the project meets its objectives.

## 7.3 Internal Review Process

The systematic review of project deliverables, based on designated assignments, is included in the CoEvolution Quality Management framework. The CoEvolution list of deliverables, along with the editors (DLs) defined in the GA and the IRs appointed by the PC, is available on the CoEvolution SharePoint under:
Documents > General > Deliverables > Deliverables_Dates&Reviewers.xlsx

IRs were chosen as scientific and technical experts either due to their expertise in the content of the deliverable or because they were not directly involved in its production. Such assignments may change depending on the availability of the partners or additional review requirements that arise during the execution of the project. The PC may further decide at his/her discretion to have any deliverable reviewed by independent external experts signing a Non-Disclosure Agreement (NDA) when necessary.

The IRs will review the deliverable and make comments for the DL. Such comments shall be made inline within the working document, using "track changes" wherever possible.

# 8 Risk Management Framework

In the following section, the Risk Management Framework (RMF) will be analysed. A systematic quality control approach for the CoEvolution project will be presented. The goal is to standardise the evaluation methods and mechanisms that will ensure a high-quality standard throughout the entire lifecycle of the project. It describes the processes for document and record management and establishes a systematic process for the identification, assessment, and treatment of potential risks. Finally, it promotes the planning and implementation of corrective actions to continuously enhance the quality of task execution and project deliverables.

## 8.1 Risk Management

Risk management is defined as an ongoing process for identifying, analysing, monitoring, controlling, mitigating, and reporting potential risks throughout the lifecycle of a project before they become issues. A “risk” refers to any potential situation that may negatively impact the project’s objectives and planned activities. Risk management within the CoEvolution project will involve the entire consortium and adhere to a structured risk management process grounded in established best practices for managing inherent and residual project risks. The Risk Management processes implemented by the CoEvolution project will be detailed, covering all stages of the risk lifecycle: Risk Identification, Risk Analysis, Risk Evaluation, Risk Treatment, Risk Communication, Risk Consultation, and Risk Reporting.

## 8.2 Roles in the risk management

### 8.2.1 Risk Manager

The RM oversees the risk management of the project, being responsible for the entire risk management process, including the monitoring and control of risks across all project activities. The RM is tasked with the development and execution of the project risk management action plan. Nonetheless, it is imperative that all partners participate, particularly the WPLs concerning risks within their respective WPs.

### 8.2.2 Risk Owners

Risk Owner is defined as a person or entity with the accountability and authority to manage a specific risk. In CoEvolution they will be designated from among the following roles: the PC, the DPC, the SPM, the various industry Partners, and the WPLs, based on their specific roles and their affinity to particular risks.

The PC has the role of Risk Owner regarding the overall coordination of project activities, the appropriateness of technical project outputs, the timely and effective achievement of milestones, and the timely production of deliverables. Each Work Package Leader will be the Risk Owner for risks pertaining to the work conducted within their respective Work Packages. All Risk Owners are responsible for the identification and management of their respective risks and must inform the Risk Manager of the status and any updates. If new risks are identified, they must be reported to the Risk Manager, and the Risk Registry dynamic template (available on the project’s repository) must be updated accordingly.

### 8.2.3 Work Package Leaders

The WPLs are responsible for identifying, reporting, and following up on implementing risks within the WPs they lead. These can be related to different project perspectives, such as deliverables, milestones, or technical evolution. Upon identification of a new risk, the WPL shall declare it in the CoEvolution Risk Registry, informing the PC, the DPC and the SPM with a proposal for an appropriate risk management strategy. Once a particular strategy has been decided for a specific risk, the WPL organises and implements the relative actions for its treatment, mitigation, or avoidance in coordination with the concerned TLs of the related WP/Task.

### 8.2.4 CoEvolution Risk Management Process

The CoEvolution Risk Management Process encompasses the following phases for managing, implementing, and controlling the project's risks (Figure 3):

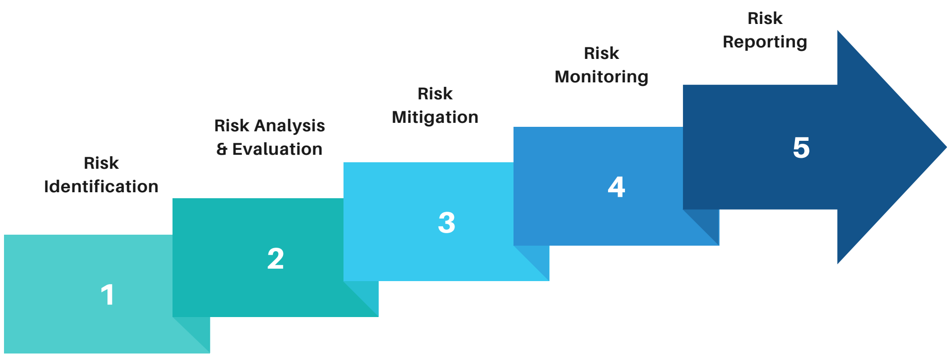



*Figure 3: Risk Phases*

**Risk Identification:** Recognizing both foreseen and unforeseen risks that could impact the project's timelines, budget, effort consumption, and outcomes.
**Risk Analysis & Evaluation:** Analysing identified risks to determine their potential impact and likelihood, thereby prioritising them based on their severity and need for immediate attention.
**Risk Mitigation:** Developing and planning specific actions to reduce the probability of occurrence or minimise the impact of prioritised risks.
**Risk Monitoring:** Continuously tracking identified risks and the effectiveness of mitigation strategies, ensuring that the risk management process is consistently applied throughout the project.
**Risk Reporting:** Ensuring effective communication about the status of risks and updates to all relevant stakeholders, including regular reporting and documentation of risk management activities.

**Phase 1**

**Risk Identification** involves recognising critical risks relative to the formulated risk management strategy. During the proposal phase of the project, a set of potential risks (foreseen risks) was acknowledged, along with corresponding mitigation strategies. These risks are documented in the Grant Agreement and registered in the EC Portal. Consequently, they have been incorporated into the initial CoEvolution Risk Registry, which serves as an internal tool for managing and monitoring project risks. However, several unforeseen risks, not included in the

Grant Agreement, may emerge from actions or events that could threaten the project's timelines, budget/effort consumption, and outcomes. Any consortium partner can identify a risk, but it is ultimately the responsibility of the WPLs to manage them. To foster an effective risk management process, identified risks can be categorised as follows:

- Risks related to the Project Management (PM).
- Risks related to the technical activities of the project (TECH).
- Risks associated with business procedures addressing the exploitation & dissemination activities of the project (BUS).
- Risks concerning the deliverables of the project (DEL).
- Risks not falling into the previous categories (OTHER).

**Phase 2**

**Risk Analysis & Evaluation** is a vital phase in project management, involving the establishment of risk scales, classification, and distribution. This stage focuses on assessing, characterising, and prioritising risks through qualitative or quantitative approaches. The objective is to pinpoint critical risks that necessitate managerial intervention.
**Roles and Responsibilities:** The process engages essential project members:

- Work Package Leader
- Project Coordinator
- Deputy Project Coordinator
- Scientific Project Manager

These team members, collaborate to categorise and rank risks, ensuring that the most crucial risks receive the necessary attention.

**Risk Classification and Prioritization:** Risks are evaluated based on two key criteria:

1. Probability of Occurrence (P)
2. Impact on the Project (I)

Both criteria are rated on a three-tier scale: 'Low', 'Medium', and 'High'.

**Risk Impact Assessment:**

- **Low Risks:** These are minor risks that might influence the success metrics of a task. The Task Leader is tasked with documenting and managing these risks and notifying the WPL.
- **Medium Risks:** These risks could potentially affect the success metrics of a specific Work Package. The WPL documents and manages these risks, informs the PC and the SPM.
- **High-Critical Risks:** These are significant risks that may severely impact the success metrics of the entire project. The PC and SPM collaboratively manage these risks.

**Risk Severity Calculation:** Following the assessment of risks based on probability and impact, the severity is calculated using a three-tier likelihood scale: 'Low', 'Medium', and 'High'. This risk severity matrix prioritises the most critical project risks, emphasising those with a high probability and/or impact.

This systematic approach ensures that project risks are thoroughly identified, assessed, and managed, thereby enhancing the overall success and robustness of the project.

| | | Probability | | |
|---|---|---|---|---|
| | | Low | Medium | High |
| Impact | Low | Low | Low | Medium |
| | Medium | Low | Medium | High |
| | High | Medium | High | High |

*Figure 4: The Risk severity scale*

**Phase 3**

**Risk Mitigation:** After assessing the risk, strategies for mitigating its impact on the project, response plans for managing the risk, or methods to completely avoid the effects of the risk are determined. This ongoing activity will extend beyond the project's lifespan. Previous decision plans may necessitate a detailed description of actions, such as gathering information and implementing mitigation measures, that should be taken either before or after the risk materializes. Consequently, a series of actions must be established according to the aspects outlined below:

1. Gather Information: Proactively collect information before the risk occurs to better understand potential threats.
2. Provide Mitigation Actions: Proactively identify strategies to eliminate or reduce the risk's probability and/or impact before it occurs.
3. Implement Contingency Plans: Reactively prepare alternative processes or backup plans to be enacted if the risk occurs, ensuring the project can continue despite disruptions.

**Phase 4**

**Risk Monitoring:** The risk monitoring phase defines the procedures for controlling risks. Partners in the CoEvolution consortium are tasked with continuously re-assessing risks and updating the CoEvolution Risk Registry at regular intervals. WPLs are responsible for informing the PC, and the SPM about the current risk status and the effectiveness of the actions taken. Actions are implemented either by the Risk Owner or other consortium members capable of addressing the identified risks. If the implemented actions are ineffective, an alternative action plan should be developed. The results of the monitoring phase are recorded in the CoEvolution Risk Registry. The risk monitoring phase includes the following activities:

1. Current Status: Review and document the status of risks (e.g., open/closed) and their severity.
2. Decision Plan Identification: Identify and adopt the relevant decision plan.
3. Effectiveness Measurement: Measure the effectiveness of the decision plan by evaluating the reduction in risk value, aiming to achieve the residual value upon completion of the decision plan.
4. Status Assessment: Assess the current risk status relative to the expected trend, including comments on the trend, required actions, and updates to the timeline.
5. Trend Exploration: Analyse risk trends for upcoming reviews.
6. New Risk Identification: Identify and assess new risks using the same process.

7. Owner Allocation: Assign owners to new risks and reallocate owners for existing risks if necessary.
8. Registry Update: Update the CoEvolution Risk Registry and communicate the changes to the consortium.

This structured approach ensures continuous monitoring, effective management, and documentation of risks, contributing to the overall success and resilience of the project.

**Phase 5**

**Risk Reporting:** The risk process and outcome are documented and reported through the prescribed mechanisms, commonly called a Risk Register. The main objectives of documenting and reporting are to:

- Communicate Risk Management Activities and Outcomes: Ensure that all the activities concerning risk management and their outcomes are communicated across an organisation in an effective way.
- Provide Information for Decision-Making: Provide all relevant information that will aid in making informed decisions.
- Improve the Risk Management Activities: Facilitate continuous improvement of risk management practices.
- Assist Interaction with Stakeholders: Assist interaction with stakeholders, more specifically those that have responsibility and accountability for risk management activities. It ensures the transparency and accountability of its risk management framework by systematic documentation and reporting of the risk management process.

## 8.3 Project-wide Risks

From the CoEvolution proposal stage onward, various risks associated with project execution have been identified along with strategies to mitigate or avoid them if they occur. These anticipated risks are documented in the Grant Agreement. The Project Coordinator, Deputy Project Coordinator, and Scientific Project Manager are responsible for monitoring and managing general unforeseen risks. They report their findings using the CoEvolution Risk Registry template.

An initial evaluation of the project-wide risks has been conducted, assessing their severity based on probability and impact, in accordance with the severity risk scale. The identified general foreseen risks are detailed in Figure 5. This thorough approach ensures effective management and mitigation of both specific and project-wide risks, supporting the overall success and resilience of the CoEvolution project.

## 8.4 Work Package & Task Related Risks

Within the WP, the WPL bears the responsibility for identifying, documenting, overseeing, and reporting risks. These risks may include technical development issues, timeline modifications, inconsistent WP objectives, inadequate WP management, underperforming partners, and other relevant challenges. The WPL is obligated to promptly inform the PC and SPM and record the risk in the CoEvolution Risk Registry.

| | Risk Identification | | | | | | | | | | Risk Assessment and Management | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Risk # | Risk Category | Short Name | Risk Description | WP1 | WP2 | WP3 | WP4 | WP5 | WP6 | Probability | Impact | Severity | Date of Identification | Risk Status | Risk Owner(s) |
| 1 | Management | MAN01 | Partner leaving the project. Allocated and possibly key-part of the work cannot be performed, delaying the project schedule | X | X | X | X | X | X | Low | Medium | Low | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL)<br>WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 2 | Management | MAN02 | Key-person left or is temporarily not available or part of work allocated to one partner cannot be performed on time and with expected quality | X | X | X | X | X | X | Low | Medium | Low | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL)<br>WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 3 | Management | MAN03 | Resources are underestimated - Part of the planned work cannot be completed in accordance with the plan | X | X | X | X | X | X | Medium | Medium | Medium | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL)<br>WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 4 | Management | MAN04 | Project schedule is partly not appropriate (e.g., due to changes) – Common project outcome and impact cannot be achieved as planned | X | X | X | X | X | X | Medium | Low | Low | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL)<br>WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 5 | Management | MAN05 | Disrupted productivity due to restraining measures | X | X | X | X | X | X | Low | Low | Low | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL)<br>WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 6 | Technical | TEC01 | Potential inaccuracies in assumed threat models, undermining the formulation of effective adversarial attacks and hindering the development of the CoEvolution framework | X | | X | X | | | Low | Low | Low | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP3 Leader (ISI)<br>WP4 Leader (TID) |
| 7 | Technical | TEC02 | Integration problems related to the software components of the CoEvolution framework and their deployment | | | X | | | X | Low | Medium | Low | 11/1/2024 | Managed | WP3 Leader(ISI)<br>WP6 Leader (AVI) |
| 8 | Technical | TEC03 | Delay in delivering the core technological components to be integrated in the platform | | | X | X | X | X | Medium | Low | Low | 11/1/2024 | Managed | WP3 Leader (ISI)<br>WP4 Leader (TID)<br>WP5 Leader (DIE)<br>WP6 Leader (AVI) |
| 9 | Technical | TEC04 | Insufficient data for assessing ML components | | | | X | X | | Low | High | Medium | 11/1/2024 | Managed | WP4 Leader (TID)<br>WP5 Leader (DIE) |
| 10 | Technical | TEC05 | Difficulty to engage with standardization bodies and WGs for the threat models and protection profiles | X | X | | | | | Medium | High | High | 11/1/2024 | Managed | WP1 Leader (ICCS)<br>WP2 Leader (CBRL) |

*Figure 5: CoEvolution Foreseen Risks*

Special meetings between the WPL, PC and SPM may be convened to plan and develop strategies aimed at mitigating, controlling, or eliminating the identified risks, thereby addressing any potential impacts on the project. The WPL is accountable for executing the agreed-upon measures, planning their implementation, monitoring the progress of the risk, and updating the CoEvolution Risk Registry as necessary.

Similarly, determining the risks connected to the pertinent Task is the responsibility of the TL. These risks could have to do with the task's technical characteristics, schedule modifications, task management, insufficient WP coordination, resource consumption, and other relevant problems. Any risks or issues that are identified must be promptly and accurately reported to the relevant WPL by the TL. In addition, the WPL, the PC and the SPM, suggest that the TL carry out the selected actions to reduce, control, or eliminate the risks that have been identified.

## 8.5 Risks Related to the Progress of the Deliverables

Early in the project, a set of risks may be identified at the deliverables level, along with appropriate steps to reduce, manage, or eliminate them. This proactive strategy aims to ensure that high-quality reports are submitted to the European Commission on schedule. The time-frame for the CoEvolution QAP deliverables has been designed with flexibility. This allows for the addition or adjustment of potential deadlines if risks arise that could affect the deliverables' timelines. Such risks include the Lead Beneficiary of the Deliverable failing to respond by the QAP-specified dates or delays caused by contributors to the Deliverables in providing their input.

# 9 Conclusions

In conclusion, this deliverable provides a comprehensive quality management plan and serves as an essential reference for the CoEvolution consortium. It offers a detailed analysis of the roles, responsibilities, and processes involved, guiding partners in adhering to established practices, guidelines, and protocols throughout the project.

The Project Manual supports the consortium in managing day-to-day activities and monitoring overall progress. By adopting the outlined methodologies, project partners can establish a shared operational framework that minimises overhead, enhances efficiency, and ensures effective collaboration. Awareness and implementation of the general aspects addressed in this document are crucial for all consortium members to contribute successfully to the project's objectives.

This manual is a cornerstone for maintaining high standards of quality and ensuring the smooth execution of CoEvolution's activities.

# Appendix A

## CoEvolution Risk Registry Template

| Risk Category | |
|---|---|
| Management | Risks related to the management of the project |
| Technical | Risks related to the technical development of the project |
| Business | Risks related to the exploitation and dissemination activities of the project |

| Risk Probability |
|---|
| Low |
| Medium |
| High |

| Risk Impact |
|---|
| Low |
| Medium |
| High |

| Risk Status | |
|---|---|
| Identified | A risk that has been identified but no action plan has been specified for it |
| Managed | A risk for which a risk management plan of actions has been specified |
| Mitigated | A risk that has been mitigated either by elimination or by transfer of liability |
| Accepted | A risk that has not been resolved but whose impact is acknowledged (along with its possible consequences) without undertaking any actions that could cause a higher cost than the risk occurrence |

**Risk Severity Scale**

| | | Probability | | |
|---|---|---|---|---|
| | | Low | Medium | High |
| Impact | Low | Low | Low | Medium |
| | Medium | Low | Medium | High |
| | High | Medium | High | High |

| Leaders | |
|---|---|
| WP1 Leader | ICCS |
| WP2 Leader | CBRL |
| WP3 Leader | ISI |
| WP4 Leader | TID |
| WP5 Leader | DIE |
| WP6 Leader | AVI |

*Figure 6: The CoEvolution Overall Risk Matrix*

| | | |
|---|---|---|
| **Risk Short Name** | | should match a risk short name in "Risks" tab |
| **Description** | | |
| **Responsible Entity** | | short name of organisation within the CoEvolution consortium which is responsible for the risk |
| **Responsible Person** | | name of the person (within the responsible entity) who is responsible for the risk |
| **Email (of responsible person)** | | email of the person (within the responsible entity) who is responsible for the risk |

| Action Types (aka Risk Mitigation Strategy) | |
|---|---|
| Information gathering | actions to get more information about the risk |
| Prevention | actions to reduce the risk probability or impact |
| Onwership Transition | actions to transfer responsibility or liability to a third party |
| Mitigation | actions to reduce the risk probability or impact |
| Acceptance | actions to transfer responsibility or liability to a third party |

| Action Status | |
|---|---|
| Planned | actions planned but not initiated yet |
| Pending | actions initiated which are yet to be completely executed |
| Completed | completed actions |

| # | Action Name (a short description) | Action Type | Action Description | Action Status | Last status update | Comments |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

*Figure 7: Risk Example*